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COSI — Naming Convention

VDW
Die Wellpappenindustrie



COSI 1.0
Corrugated Standard Interface

Open industry standard for machine data exchange
in corrugated cardboard manufacturing

Imprint

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VDW (Association of the German Corrugated Cardboard Industry e.V.)

Authors

Michael Sachs

Robert Schmid

Daniel Malolepszy

Dennis Strasser

Tobias Winkler

Technical editors

Technical specification development and quality assurance.

André Seifert (ePS)

Robin Boeckxstaens (OMP)

Contact

Christiane Trumpp

transparente-daten@vdw-da.de

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The PDF edition is a reproduction of the Markdown source published in the official COSI GitHub repository. In the event of any discrepancy, the Markdown source for the corresponding document version shall prevail.

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1 Case Style

All JSON field names use **camelCase**.

```
machineId       [OK]
machine_id      [NO]
MachineId       [NO]
```

2 Language

All field names are **English**. Domain terms originating from German or other languages must be translated to their English equivalent before use as a field name.

3 Abbreviations

Avoid abbreviations unless the abbreviated form is the universally recognised standard in the corrugated manufacturing or industrial automation domain.

Permitted abbreviations:

Abbreviation	Meaning
`uom`	Unit of measure
`hbt`	Heartbeat
`cos`	Change of status
`lm`	Linear metres
`pH`	Potential of hydrogen
`dE2000`	Delta E 2000 colour difference
`OEE`	Overall Equipment Effectiveness
`FSC`	Forest Stewardship Council certification
`gsm`	Grams per square metre — paper grammage
`ID`	Identifier — envelope-level only (see Section 4)

4 Identifier Suffixes

`ID` — Envelope-level globally unique identifiers

Reserved for the two envelope fields that carry globally unique message identifiers:

```
"messageID":     "uuid",
"correlationID": "uuid"
```

`Id` — Other identifiers

All other identifier fields use lowercase `Id`:

```
"machineId", "colorId", "statusId", "streamId", "schemaId", "connectionId"
```

`connectionId` is globally unique despite its `Id` spelling; only the two established message identifiers use `ID`. See Message Properties.

`Ref` — Cross-system references

Fields that reference an entity managed in an external system (ERP, MES, machine) use the `Ref` suffix. Their requiredness and permission to use null are field-specific; follow the payload schema.

Field	Meaning
`jobRef`	Manufacturing order reference
`articleRef`	Article / product specification reference
`taskRef`	Manufacturing step reference within the order
`setupRef`	Corrugator quality reference to setupSettings.setupId within the same site/machine context

5 Boolean Fields

Boolean fields use a descriptive adjective. The `is*` prefix is not used.

```
active          [OK]
crossFlute      [OK]
autoStopFeed    [OK]
isActive        [NO]
```

6 Units of Measure

Field name

The unit of measure field is always named **`uom`**. The name `unit` is not used — it is ambiguous with production unit, pallet unit, and similar domain concepts.

Per-value declaration

Every field that carries a measured quantity includes its own `uom`. There is no global unit declaration at the message or envelope level. Messages are self-describing.

Unit system

v1.0 is metric only. Use the unit symbols and field-specific tokens defined by the schemas and this document. Existing tokens such as `pcs`, `stacks`, `cuts`, `cycles`, `dpi`, `station`, `FEFCO` and `dE2000` retain their specified uses; not every token is an SI unit symbol. Imperial measurement alternatives (`in`, `ft/min`, `°F`, `BTU`, `psi`, …) are **not** valid in v1.0 — producers convert those measurements to metric before publishing. General imperial-unit support is deferred to v1.1. See ADR-000013 and the existing fixed-token corrections in ADR-000016.

Numeric precision

Numeric precision follows the data type. Producers send the precision their source data type provides; consumers must not assume a fixed number of decimal places. COSI imposes no additional precision restrictions or limits.

Preferred `uom` values

Quantity	uom value
Length	mm or m
Area	m²
Weight	t (metric tonne)
Grammage (gsm)	g/m²
Speed — converting	pcs/h
Speed — corrugator	m/min
Temperature	°C
Humidity	%RH
Energy	Wh
Steam	kg
Water / Air volume	m³
Colour accuracy	dE2000
Ink viscosity	s (seconds, Ford cup)
Force / Pressure	kN
Print resolution	dpi

7 Quality Check Shape

Quality checks use the common structure described in Message Properties — Quality Check Shape. Field naming, enum casing and null handling follow the conventions in this document.

8 Enum Casing Rules

The default enum style is **lowercase**, using **hyphens** for multi-word values. Preserve the exact case of values defined for a particular field.

Existing exceptions: check/setup-response results use OK / NOK; failure-mode codes use the uppercase values in the fault-code table (ADR-000015). Unit symbols and classification tokens retain their defined spelling (for example Wh, g/m², FEFCO). Suggested values for open enums retain their documented spelling; Section 10 does not restrict them to a closed list.

For the full value lists and descriptions, see `cosi-message-properties-v1.0.pdf`.

9 Arrays vs Objects

Use	When
Array	Variable-length, positional, or repeating items (stream items, colours, periphery devices)
Object	Named, fixed sets of fields (quality check sections, machine settings groups)

10 Open vs Closed Enumerations

Some fields have a defined set of known values but must remain extensible to allow industry-specific or future additions without breaking validation.

Type	When to use	Schema implementation
Closed enum	The value set is complete and fixed by the standard (e.g., machine status values, check result OK/NOK, failure severity, failure mode code)	`"enum": [...]` — validation rejects unknown values

Type	When to use	Schema implementation
Open enum	The value set has well-known suggestions but producers may send additional values (e.g., FSC certification level, glue style, discharge direction)	`"type": "string"` with suggested values listed in `description` — validation accepts any string

Suggested values for open enums must be documented in the schema `description` field, not enforced via e
`num`.

11 Null vs Absent

- Required fields remain required. Use `null` only where that specific field permits it.
- Optional unavailable or inapplicable measurements may be omitted where appropriate.
- Zero is a reported zero, not a substitute for an unavailable value.
- Preserve the specified inactive-device conventions (for example zero positions for unused scoring knives).
- An empty or absent set of quality checks does not mean the material is OK.

12 Qualified Value Groups

Measurements that carry multiple temporal or contextual variants of the same physical value are represented as a **grouped object**. The group name is the measurement concept; the qualifier is the key within the group; u
`om` is declared once at the group level.

The vocabulary below explains qualifier names; it does not authorize all qualifiers in every payload or determine message triggers. Each schema defines the permitted fields, requiredness and null values. Setup carries settings, not production actuals.

Qualifier Vocabulary

Qualifier	Meaning	Typical message type
`target`	Reference or goal value — best practice or job-specific aim	HBT, COS
`scheduled`	Value committed in the production plan (from ERP / MES)	COS
`actual`	Final measured or achieved value for a completed period	COS
`current`	Reading at the original event/observation time	HBT
`average`	Computed mean over the reporting period	HBT, COS
`minimum`	Minimum value reached over the reporting period	COS
`maximum`	Maximum value reached over the reporting period	COS

Example from the performance payload

The `speed` object below is taken from the performance example. Its shared `uom` applies to both readings. The performance schema requires `current` and `uom`, permits `target`, and permits no other properties in this object.

```
{
  "speed": {
    "uom": "m/min",
    "current": 182,
    "target": 200
  }
}
```

This is a payload fragment, not a complete message.

Rules

- Include only schema-permitted qualifiers; omit an inapplicable qualifier only when optional.
- Use `null` for an unavailable qualifier only where its schema permits null; do not omit required fields.
- All values within a group share the same `uom`. If two variants of the same measurement require different units, they must be modelled as separate fields.

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